MR. ALLEN'S 8TH GRADE MATH

NC Check-In Benchmark Review

Notes • Benchmark-Style Practice • Test Strategies

NS.1 EE.1 EE.7 F.1 G.3

Name _____ Date _____ Block _____

What's on this Check-In

Standard	Topic & "I can" goal	In this packet
NC.8.NS.1	**Rational & Irrational Numbers** I can tell rational from irrational numbers and write a repeating decimal as a fraction.	Review
NC.8.EE.1	**Integer Exponents** I can use exponent rules (including zero and negative exponents) to write equivalent expressions.	Review
NC.8.EE.7	**Solving Linear Equations & Inequalities** I can solve multi-step equations and inequalities and tell if an equation has one, none, or infinitely many solutions.	Review
NC.8.F.1	**What Is a Function?** I can decide if a table, graph, mapping diagram, or set of ordered pairs is a function.	New • Full Notes
NC.8.G.3	**Transformations with Coordinates** I can describe and find the coordinates after a translation, reflection, rotation, or dilation.	New • Full Notes

How to use this packet

- **Page 2** — Test strategies (read these first, every time).
- **Notes** — Quick refreshers for NS.1, EE.1, EE.7; full notes for F.1 and G.3.
- **Practice** — 20 Check-In style questions, 4 per standard, then a self-check.

Question types you'll see

- (A) **Multiple Choice** — pick ONE.
- [A] **Select All That Apply** — could be 2, 3, or more.
- [ ] **Numeric Entry** — type your own answer.

Strategies

Math Test-Taking Strategies

Review every time

1 Box the question.
Before any math, draw a box around what they are actually asking for. Is it x? The **image** of point B? The number of solutions?

2 Hunt for trap words.
Circle **NOT**, **EXCEPT**, **greatest**, **least**, **all**. One missed "NOT" flips the whole answer.

3 Plug it back in.
Got $x = 5$? Put 5 into the ORIGINAL equation. Both sides match → you're right. Works on Multiple Choice too: test each choice.

4 Eliminate first.
Cross out choices that can't be right (wrong sign, wrong quadrant, too big). Two left is a 50/50 — much better than 25%.

5 Select All = True/False.
Judge every option on its own: **T** or **F**. Don't stop after one. Expect 2 or more correct.

6 Numeric Entry has no safety net.
No choices to compare, so check it twice. Watch negative signs, simplify fractions, and give the form asked for (fraction vs. decimal).

7 Draw it.
Sketch a quick grid, mapping diagram, or number line on scratch paper. Seeing it beats picturing it — especially for G.3 and F.1.

8 Estimate, then check.
Make a quick guess before solving. If your final answer is way off from your estimate, go back and look.

9 Wrong answers are built from mistakes.
Test writers put common errors in the choices (added exponents instead of multiplying, rotated the wrong way). If your answer is there, still double-check it.

10 Never leave one blank.
There is no penalty for guessing. Eliminate what you can, then make your best pick and move on.

C.U.B.E.S. for Word Problems

Circle key numbers **U**nderline the question **B**ox math words **E**valuate & eliminate **S**olve & check

Watch-Out List — the mistakes that cost points

Standard	Common mistake	Fix
NS.1	Thinking 0.272727… is irrational because it "never ends."	Repeating = rational. Irrational means it never ends **and** never repeats.
EE.1	Multiplying the bases: $3^2 \cdot 3^4 = 9^6$. Or thinking a negative exponent makes a negative number.	Same base stays the same base. Negative exponent = flip to the other side of the fraction.
EE.7	Distributing to only the first term; forgetting to flip the sign.	Draw the arrows to EVERY term. Multiply/divide by a negative → flip the inequality.
F.1	Saying "not a function" because a **y**-value repeats.	Only a repeated **x** (input) with different outputs breaks a function.
G.3	Mixing up the x-axis and y-axis reflections; rotating clockwise when it says counterclockwise.	Use the rule chart. Reflect over x-axis → **y** changes sign.

NS.1 Rational & Irrational Numbers

Quick Refresher

I can tell rational from irrational numbers and write a repeating decimal as a fraction.

Rational

Can be written as a fraction $\frac{a}{b}$ of integers ($b \neq 0$).

Its decimal **ends** or **repeats**.

$-7 \quad \frac{3}{8} \quad 0.25 \quad 0.\overline{3} \quad \sqrt{49} = 7 \quad \sqrt{0.25} = 0.5$

Irrational

CANNOT be written as a fraction.

Its decimal **never ends AND never repeats**.

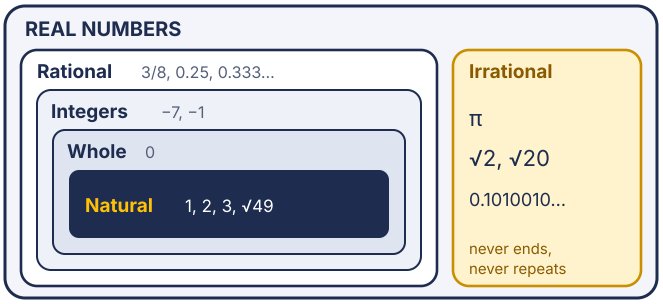
$\pi \quad \sqrt{2} \quad \sqrt{20} \quad 0.1010010001\ldots$

√ of a whole number that is NOT a perfect square → irrational.

Where numbers live

Every real number is **either** rational **or** irrational — never both. Integers, whole numbers, and natural numbers are all **inside** the rational box.

√49 = 7, so it's a natural number — even with a square root sign!



Repeating decimal → fraction: the 9's Rule

1. **Count** the repeating digits.
2. **Denominator** = that many 9's (one 9 for each repeating digit).
3. **Numerator** = the repeating digits.
4. **Simplify.**

$0.\overline{27} = 0.272727\ldots$

2 digits repeat → 99

$\frac{27}{99} = \frac{\mathbf{3}}{\mathbf{11}}$ $(\div 9)$

Repeating digits	Denominator	Example
1 digit	9	$0.\overline{3} = \frac{3}{9} = \frac{1}{3}$
2 digits	99	$0.\overline{45} = \frac{45}{99} = \frac{5}{11}$
3 digits	999	$0.\overline{125} = \frac{125}{999}$

Remember: the 9's rule works when the decimal starts repeating right after the decimal point. Always simplify your fraction.

EE.1 Integer Exponents

Quick Refresher

I can use the properties of integer exponents to write equivalent expressions.

Rule	In words	Example
Product $a^m \cdot a^n = a^{m+n}$	Same base, multiplying → **add** exponents	$2^3 \cdot 2^5 = 2^8$
Quotient $a^m \div a^n = a^{m-n}$	Same base, dividing → **subtract** (top − bottom)	$5^7 \div 5^9 = 5^{-2}$
Power of a power $(a^m)^n = a^{mn}$	Exponent outside parentheses → **multiply**	$(x^4)^3 = x^{12}$
Power of a product $(ab)^n = a^n b^n$	Give the exponent to **every** factor inside	$(2y^3)^2 = 4y^6$
Zero $a^0 = 1$	Anything (not 0) to the zero power is 1	$(-8)^0 = 1$
Negative $a^{-n} = \frac{1}{a^n}$	Negative exponent → **flip** it across the fraction bar. It does NOT make the number negative.	$4^{-2} = \frac{1}{16}$

Why do we ADD exponents?

An exponent counts how many times the base is multiplied. Put the groups together and count.



2^3 (3 twos) · 2^2 (2 twos) = 2^5 — count the twos: 3 + 2 = 5

Careful: $(-4)^2 = 16$ but $-4^2 = -16$. The negative is only inside the power if it's inside the parentheses. Final answers usually have **positive exponents only**.

EE.7 Solving Linear Equations & Inequalities

Quick Refresher

I can solve multi-step equations and inequalities (variables on both sides) and decide how many solutions an equation has.

The 4-step game plan

1. **Distribute** to clear parentheses.
2. **Combine like terms** on each side.
3. **Collect** variables on one side (inverse operations).
4. **Isolate** the variable. Then **check**.

Fractions? Multiply EVERY term by the LCD to clear them first.

$3(x - 4) + 5 = x + 9$

$3x - 12 + 5 = x + 9$

$3x - 7 = x + 9$

$2x - 7 = 9$

$2x = 16 \rightarrow \mathbf{x = 8}$

Check: $3(4)+5 = 17$ and $8+9 = 17$ ✓

How many solutions? — look at what's left when the variables cancel

One solution	No solution	Infinitely many solutions
You end with **x = a number**.	Variables cancel, you get a **false** statement.	Variables cancel, you get a **true** statement.
$4x + 1 = 2x + 7$ $x = 3$	$2(x + 3) = 2x + 5$ $6 = 5$ ×	$3(x + 2) = 3x + 6$ $6 = 6$ ✓
The x's do not cancel.	The x's cancel and you get something false.	Both sides are exactly the same

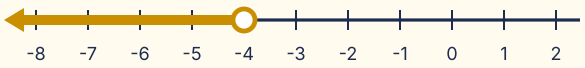
Inequalities: same steps + ONE rule

Multiply or divide both sides by a **negative** → **flip the sign**.

$-3x + 7 > 19$

$-3x > 12$

$x < -4$ (÷ −3 → flip)



Open circle: −4 is NOT included (<). Closed circle for ≤ or ≥.

Word problems → read the context

"At most" → ≤ "at least" → ≥

If the answer must be a whole number (people, tickets, shirts), **round to what makes sense** — usually down for "can afford."

F.1 What Is a Function?

Full Notes — New

I can explain that a function gives each input exactly one output, and decide if a table, graph, mapping diagram, or list of ordered pairs is a function.

Definition

A **function** is a rule that assigns to each **input** (x) **exactly one output** (y). The graph of a function is the set of ordered pairs (input, output).

Think of a vending machine: press B4 → you always get the same snack. If B4 sometimes gave chips and sometimes gave cookies, the machine would be broken — **not a function**.

Vocabulary

Relation	Any set of ordered pairs.
Input	The x-values (what goes in).
Output	The y-values (what comes out).
Function	A relation where no input has two different outputs.

The ONE question to ask

"Does any **x** repeat with a **different y**?"

- **Yes** → NOT a function.
- **No** → function.
- Repeated **y**-values are fine! Two kids can share a birthday.

Four ways to see it

1. Table

x	y
1	3
2	3
3	5

Function ✓
y repeats — OK

x	y
2	1
4	6
2	8

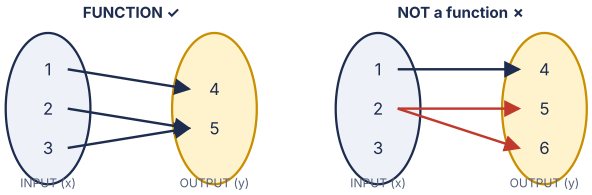
NOT ×
x = 2 → 1 and 8

2. Ordered pairs

{(0, 2), (1, 4), (2, 6)}
Function ✓ — every x is different

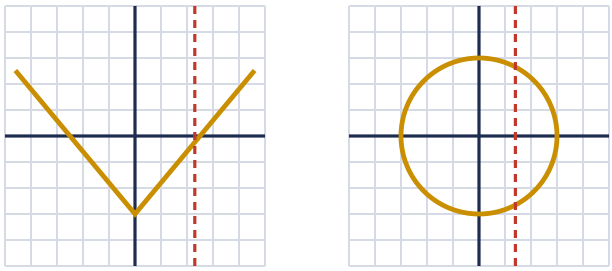
{(5, 1), (3, 2), (3, 7)}
NOT × — x = 3 has two outputs

3. Mapping diagram



Count arrows leaving each input: more than one arrow from the same input → NOT a function.

4. Graph — Vertical Line Test



Function ✓ — hits once **NOT ×** — hits twice

Slide a vertical line across. If it ever touches the graph in 2+ points → NOT a function.

Functions in real life

- **Function:** input = number of $8 tickets, output = total cost. Each number of tickets has one cost.
- **Function:** input = a student, output = their birthday.
- **NOT:** input = an age, output = a student with that age. Age 13 could match many students.

Benchmark trick: "Which value of *k* makes the relation NOT a function?" Look for the choice that **matches an x-value already used** with a different y. A vertical line (x = 2) is never a function; a horizontal line (y = 2) always is.

G.3 Transformations with Coordinates

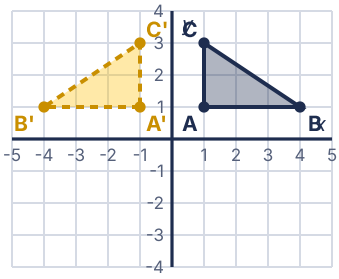
Full Notes — New

I can move a shape on a grid and give the new points.

Vocabulary

Pre-image	The original figure: A, B, C
Image	The new figure, named with **primes**: A′, B′, C′
Rigid motion	Moves a figure but keeps its size & shape → **congruent** (≅)
Dilation	Stretches or shrinks → **similar** (~), not congruent
Scale factor (k)	k > 1 enlarges; 0 < k < 1 shrinks

Example: reflect △ABC over the y-axis



A(1, 1) → A′(−1, 1) B(4, 1) → B′(−4, 1) C(1, 3) → C′(−1, 3)

x changes sign, y stays the same.

The Rule Chart — know these cold

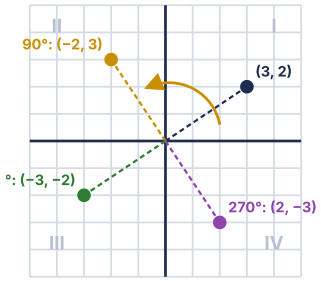
Transformation	Coordinate rule	Example: (3, −2)	How to remember
Translation a units right/left, b units up/down	$(x, y) \to (x + a, y + b)$	4 left, 1 up: $(-1, -1)$	Right/left changes x. Up/down changes y. Left & down are negative.
Reflection over the x-axis	$(x, y) \to (x, -y)$	$(3, 2)$	Flip over the x-axis → **y** flips sign.
Reflection over the y-axis	$(x, y) \to (-x, y)$	$(-3, -2)$	Flip over the y-axis → **x** flips sign.
Rotation 90° counterclockwise (same as 270° clockwise)	$(x, y) \to (-y, x)$	$(2, 3)$	Swap, then make the **new first** number opposite.
Rotation 180° (either direction)	$(x, y) \to (-x, -y)$	$(-3, 2)$	Both signs flip. No swap.
Rotation 270° counterclockwise (same as 90° clockwise)	$(x, y) \to (y, -x)$	$(-2, -3)$	Swap, then make the **new second** number opposite.
Dilation by scale factor k, centered at origin	$(x, y) \to (kx, ky)$	$k = 2$: $(6, -4)$	Multiply **both** coordinates by k. Side lengths × k, angles stay the same.

What stays the same?

	Size	Shape & angles	Congruent?
Translate / Reflect / Rotate	✓	✓	**Yes**
Dilate	×	✓	No — **similar**

Quick-check tricks

- **Quadrant check:** a 90° rotation moves a point one quadrant over; 180° moves it to the opposite quadrant.
- **Counterclockwise** = the direction of the quadrant numbers I → II → III → IV.
- Turn your paper! Rotate your scratch sketch to see a rotation.



NS.1 Practice — Rational & Irrational Numbers

4 questions

1 NC.8.NS.1 Select All That Apply

Select **ALL** of the numbers that are **irrational**.

A $\sqrt{49}$

B $\sqrt{20}$

C $0.\overline{37}$

D π

E $-\frac{5}{8}$

F 0.1010010001... (pattern: one more 0 each time)

Show your work

2 NC.8.NS.1 Numeric Entry

Use the 9's rule. Write the repeating decimal $0.\overline{36}$ = 0.363636... as a fraction in simplest form.

Answer:

Show your work

3 NC.8.NS.1 Multiple Choice

Which fraction is equal to $0.\overline{27}$ = 0.272727... ?

A $\frac{27}{100}$

B $\frac{3}{11}$

C $\frac{2}{7}$

D $\frac{27}{90}$

Show your work

4 NC.8.NS.1 Multiple Choice

Which statement is true?

A Every fraction has a decimal that terminates (ends).

B 0.161616... is irrational because it never ends.

C $\sqrt{50}$ is irrational because 50 is not a perfect square.

D π is rational because $\pi = \frac{22}{7}$.

Show your work

EE.1 Practice — Integer Exponents

4 questions

5 NC.8.EE.1 Multiple Choice

Which value is equal to the expression?

$$\frac{3^4 \cdot 3^{-6}}{3^2}$$

(A) -81

(B) $\frac{1}{81}$

(C) $\frac{1}{9}$

(D) 81

Show your work

6 NC.8.EE.1 Numeric Entry

What is the value of the expression?

$$(2^3)^2 \cdot 2^{-4} \cdot 7^0$$

Answer:

Show your work

7 NC.8.EE.1 Select All That Apply

Select **ALL** expressions that are equal to $\frac{1}{16}$.

[A] 4^{-2}

[B] 2^{-4}

[C] -16

[D] $(-4)^{-2}$

[E] -4^{-2}

[F] $\frac{2^8}{2^4}$

Show your work

8 NC.8.EE.1 Multiple Choice

Which expression is equivalent to the expression below?

$$\frac{(x^5y^{-2})^3}{x^2y^{-4}}$$

(A) $\frac{x^{13}}{y^2}$

(B) $x^{13}y^2$

(C) $\frac{x^6}{y^8}$

(D) $\frac{x^{17}}{y^{10}}$

Show your work

EE.7 Practice — Equations & Inequalities

4 questions

9 NC.8.EE.7 Numeric Entry

Solve for *x*.

$$4(x - 3) + 2x = 2x + 8$$

Answer:

Show your work

10 NC.8.EE.7 Multiple Choice

How many solutions does this equation have?

$$2(3x - 5) + 4 = 6x - 6$$

(A) No solution

(B) Exactly one solution, x = 0

(C) Exactly one solution, x = −6

(D) Infinitely many solutions

Show your work

11 NC.8.EE.7 Numeric Entry

Solve for x.

$$\frac{2}{3}x - 4 = \frac{1}{6}x + 5$$

Answer: _____

Show your work

12 NC.8.EE.7 Multiple Choice

Jordan has $85. He buys a jacket for $34 and wants to spend the rest on T-shirts that cost $12 each. What is the **greatest** number of T-shirts Jordan can buy?

(A) 4
(B) 4.25
(C) 5
(D) 7

Show your work

F.1 Practice — Functions

4 questions

13 NC.8.F.1 Multiple Choice

Which table represents y as a function of x?

(A)

x	1	2	1	3
y	4	5	6	7

(B)

x	2	4	6	8
y	5	5	5	5

(C)

x	-1	0	-1	2
y	3	3	4	5

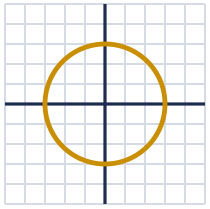
(D)

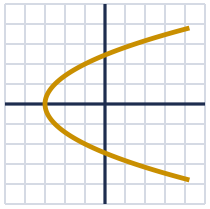
x	5	5	5	5
y	1	2	3	4

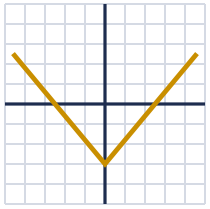
Show your work

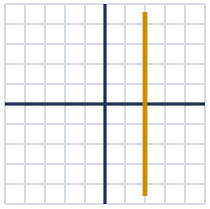
14 NC.8.F.1 Multiple Choice

Which graph represents a function?

(A) 

(B) 

(C) 

(D) 

Show your work

15 NC.8.F.1 Select All That Apply

Select **ALL** of the relations that are functions.

A {(1, 2), (2, 3), (3, 4)}

B {(2, 1), (2, 3), (4, 5)}

C {(−3, 5), (0, 5), (3, 5)}

D {(0, 0), (1, 1), (1, −1)}

E {(7, 2), (8, 2), (9, 3)}

Show your work

16 NC.8.F.1 Multiple Choice

The relation below is shown as a set of ordered pairs.

$$\{(2, 5), (4, 7), (k, 9), (6, 11)\}$$

Which value of k would make the relation **NOT** a function?

(A) 1

(B) 3

(C) 4

(D) 8

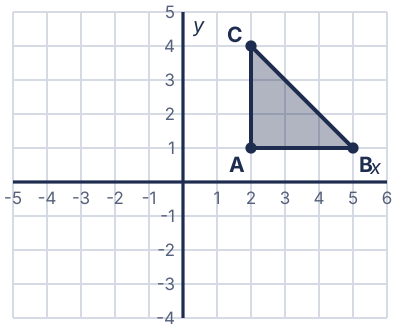
Show your work

G.3 Practice — Transformations

4 questions

17 NC.8.G.3 Multiple Choice

Triangle ABC is translated **4 units left** and **3 units down** to form triangle A′B′C′. What are the coordinates of **B′**?

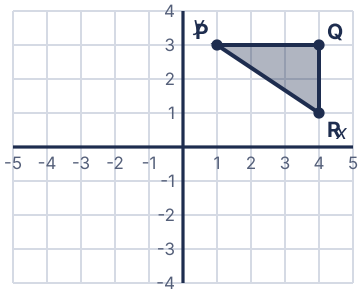


(A) (1, −2)
(B) (9, −2)
(C) (1, 4)
(D) (−2, −2)

Show your work

18 NC.8.G.3 Multiple Choice

Triangle PQR is reflected across the **y-axis**. Which set of points are the vertices of P′Q′R′?

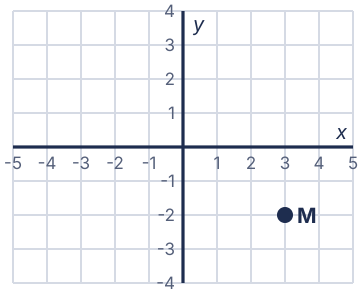


(A) P′(1, −3), Q′(4, −3), R′(4, −1)
(B) P′(−1, 3), Q′(−4, 3), R′(−4, 1)
(C) P′(−1, −3), Q′(−4, −3), R′(−4, −1)
(D) P′(3, 1), Q′(3, 4), R′(1, 4)

Show your work

19 NC.8.G.3 Multiple Choice

Point M(3, −2) is rotated **90° counterclockwise** about the origin. What are the coordinates of M′?

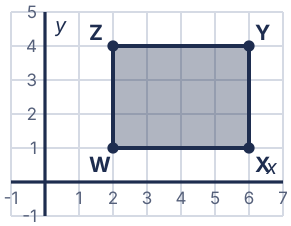


(A) (2, 3)
(B) (−2, −3)
(C) (−3, 2)
(D) (3, 2)

Show your work

20 NC.8.G.3 Select All That Apply

Rectangle WXYZ is dilated by a scale factor of $\frac{1}{2}$ centered at the origin. Select **ALL** true statements.



(A) W′ is at (1, 0.5).
(B) Z′ is at (4, 8).
(C) The image has side lengths 2 and 1.5.
(D) The image is congruent to WXYZ.
(E) Every angle of the image still measures 90°.

Show your work

Score 5 Practice — Explain Your Thinking

2 questions

21 NC.8.EE.1 Multiple Choice

Score 5. Which shows **why** $6^0 = 1$?

(A) $6 \times 0 = 0$, and $0 + 1 = 1$.
(B) $6^3 \div 6^3 = 6^0$, and a number divided by itself is 1.
(C) $6^1 = 6$, so $6^0 = 6 - 5 = 1$.
(D) It is only a rule to memorize. There is no reason.

Show your work

22 NC.8.G.3 Multiple Choice

Score 5. Triangle T is reflected over the x-axis and then dilated by a scale factor of 2 to make triangle T′. Which statement is true?

(A) T and T′ are congruent, because a reflection is a rigid motion.

(B) T and T′ are similar but not congruent. Each side of T′ is 2 times as long.

(C) T and T′ are similar. Each angle of T′ is 2 times as large.

(D) T and T′ are neither similar nor congruent.

Show your work

Self-check: how did I do?

Standard	Questions	My score	Next step
NS.1	1–4	____ / 4	4/4 — Ready! 3/4 — Re-read the notes page. 0–2/4 — See your teacher for a re-teach.
EE.1	5–8	____ / 4	
EE.7	9–12	____ / 4	
F.1	13–16	____ / 4	
G.3	17–20	____ / 4	
Score 5	21–22	____ / 2	These ask you to explain WHY.